SIGHTLINE

CASE DEMO

RECONSTRUCTION ANALYSIS REPORT

Car into turning tractor-semitrailer at an uncontrolled median crossover, US-27A near Williston, Florida (2016)

National Transportation Safety Board, Highway Accident Report (car and tractor-semitrailer collision near Williston, Florida, May 7, 2016), NTSB/HAR-17/02 (PB2017-102600), adopted September 12, 2017. Accident ID HWY16FH018.
`https://www.ntsb.gov/investigations/AccidentReports/Reports/HAR1702.pdf`
`https://www.ntsb.gov/investigations/Pages/HWY16FH018.aspx`

CASE REFERENCE	NTSB HWY16FH018 — US-27A, Williston, Florida, May 7, 2016
OVERALL RESULT	contradiction found · 5 findings (1 contradiction, 3 consistent, 1 unverifiable)
RECONSTRUCTION	world_v2 (version 2), 2 review rounds
VALIDATION	11 of 12 published facts reproduced within tolerance (NTSB HAR-17/02)
RUN MODE	mock — cached model outputs replayed; no live model calls were made for this report
REPORT GENERATED	October 25, 2026 at 12:29 PM PDT

ANALYSIS DRAFT FOR EXPERT REVIEW — NOT EVIDENCE

This document is a working analysis prepared to help a qualified accident-reconstruction expert review the testimony in this case. It is not an expert opinion and is not offered as evidence. Every value in the reconstruction is tied to a citation; values marked *inferred* or *assumed* are modeling choices that an expert must confirm.

Kinematic reconstruction: motion fitted to cited evidence; not a crash-physics simulation.

SIMULATED DEPOSITION / FICTIONAL WITNESSES

The physical facts of this case are taken from the published investigation (NTSB HWY16FH018). The witnesses, their statements and the videotaped deposition are **fictional** and were created to demonstrate the method. No real person's testimony is analysed here.

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1 Summary of findings

NO.	STATUS	FINDING	SEVERITY
1	CONTRADICTION	Witness B could not have watched the car up to impact from where she marked she was standing: Vehicle T's tractor, then its semitrailer, block her view from about t=9.1s.	high
2	CONSISTENT	Her account of seeing the car come over the crest and down the grade is physically possible: the view from her mark is clear until about t=9s.	medium
3	CONSISTENT	Witness A had a clear line of sight to the passenger sedan at t=2s, consistent with the statement.	low
4	CONSISTENT	Witness C had a clear line of sight to the truck-tractor at t=5s, consistent with the statement.	low
5	UNVERIFIABLE	No skid evidence to test: NTSB found no pre-impact tire marks and no braking.	low

2 Evidence received

Sizes and running times are read from the files as stored. Hashes are SHA-256 of the file given to the model (first 16 hex digits; full values in Section 8).

EXH.	FILE	TYPE	SIZE	LENGTH	SHA-256
E-1	deposition.mp4 Simulated deposition (fictional witness)	Video	4.0 MB 4,244,843 bytes	30:02 1,801.8 s	8811f02e4e9e62da
E-2	deposition.srt	Document	30.0 KB 30,738 bytes	—	4c0cb2f766810599
E-3	facts.md	Document	6.7 KB 6,883 bytes	—	16a2ac328322655c
E-4	map.png	Image	53.3 KB 54,576 bytes	—	e44f99efb5dd4830
E-5	police_report.md	Document	6.4 KB 6,532 bytes	—	6b994577e18a9041
E-6	witness_A.md	Document	1.7 KB 1,729 bytes	—	57b80e791b764249
E-7	witness_C.md	Document	1.5 KB 1,507 bytes	—	14873e9544ba08b9
E-8	deposition_transcript.txt	Document	20.2 KB 20,708 bytes	—	416d65e03cfb46ae

3 Findings

Finding 1 CONTRADICTION

SEVERITY: HIGH · REF F1

Witness B could not have watched the car up to impact from where she marked she was standing: Vehicle T's tractor, then its semitrailer, block her view from about t=9.1s.

Her eye-level render at t=9.9s shows the tractor filling the direction of the approaching car; the overhead render shows the truck crossing between her X and the eastbound lanes. The ray-cast agrees (blocked by veh_T_trailer, 28.3 m to the car). She testified "I had a clear view of the car the whole way in. I watched it right up until the moment it hit. Nothing was in the way." (`Depo 27:46`).

EVIDENCE CITED

1. deposition.mp4 — `Depo 27:04` (position marked on Exhibit 4)
2. deposition.mp4 — `Depo 27:46` (claim)
3. deposition.mp4 — `Depo 28:17` (repeats: until it went under the trailer)
4. NTSB/HAR-17/02 via facts.md (F19) — `p.1` , `p.21`
5. NTSB/HAR-17/02 via facts.md (F30) — `p.1–2` , `p.29–30`
6. Render — wit_B.png @ t=9.9s
7. Independent check — sightline wit_B→veh_A from marked position @ t=9.9s: blocked by veh_T_trailer

INDEPENDENT CHECK RESULTS

CHECK	RESULT	FINDING OF THE CHECK
Line of sight wit_B	BLOCKED	Line of sight wit_B → veh_A at t = 9.90 s: **Blocked by veh_T_trailer** (target 28.3 m away). Robustness: blocked from **25 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — robust (veh_T_trailer 15, veh_T_tractor 10). Gaze: target **28.5°** off the line of gaze (gaze assumed toward mark_impact); not peripheral (30° or less); camera FOV 60°.

Finding 2 CONSISTENT

SEVERITY: MEDIUM · REF F2

Her account of seeing the car come over the crest and down the grade is physically possible: the view from her mark is clear until about t=9s.

EVIDENCE CITED

1. deposition.mp4 — `Depo 14:57`
2. NTSB/HAR-17/02 via facts.md (F28) — `p.29`
3. Independent check — sightline wit_B→veh_A from marked position @ t=8.5s: clear

INDEPENDENT CHECK RESULTS

CHECK	RESULT	FINDING OF THE CHECK
Line of sight wit_B	BLOCKED	Line of sight wit_B → veh_A at t = 9.90 s: **Blocked by veh_T_trailer** (target 28.3 m away). Robustness: blocked from **25 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — robust (veh_T_trailer 15, veh_T_tractor 10). Gaze: target **28.5°** off the line of gaze (gaze assumed toward mark_impact); not peripheral (30° or less); camera FOV 60°.

Finding 3 CONSISTENT

SEVERITY: LOW · REF F3

Witness A had a clear line of sight to the passenger sedan at t=2s, consistent with the statement.

EVIDENCE CITED

1. witness_A.md — written statement
2. Independent check — sightline wit_A→veh_A @ t=2s: clear

INDEPENDENT CHECK RESULTS

CHECK	RESULT	FINDING OF THE CHECK
Line of sight wit_A	CLEAR	Line of sight wit_A → veh_A at t = 2.00 s: **Clear** (target 639.5 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **2.4°** off the line of gaze (gaze assumed toward mark_impact); not peripheral (30° or less); camera FOV 60°.

Finding 4 CONSISTENT

SEVERITY: LOW · REF F4

Witness C had a clear line of sight to the truck-tractor at t=5s, consistent with the statement.

EVIDENCE CITED

1. witness_C.md — written statement
2. Independent check — sightline wit_C→veh_T_tractor @ t=5s: clear

INDEPENDENT CHECK RESULTS

CHECK	RESULT	FINDING OF THE CHECK
Line of sight wit_C	CLEAR	Line of sight wit_C → veh_T_tractor at t = 5.00 s: **Clear** (target 29.2 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **17.1°** off the line of gaze (gaze assumed toward mark_impact); not peripheral (30° or less); camera FOV 60°.

Finding 5 UNVERIFIABLE

SEVERITY: LOW · REF F5

No skid evidence to test: NTSB found no pre-impact tire marks and no braking.

EVIDENCE CITED

1. NTSB/HAR-17/02 via facts.md (F22) — `p.5`
2. NTSB/HAR-17/02 via facts.md (F18) — `p.15`

INDEPENDENT CHECK RESULTS

CHECK	RESULT	FINDING OF THE CHECK
Perception-reaction time veh_A	SUFFICIENT	Perception-reaction time: veh_A had **10.43 s** from first sight of veh_T_trailer to impact (first visible t = 0.00 s); PRT range 1.0–2.5 s, typical 1.5 s — **sufficient**. Time from the conflict first arising to impact: 2.88 s.

CHECK	RESULT	FINDING OF THE CHECK
Plausibility veh_A	PASS	Kinematic plausibility: peak 0.02 g at t = 4.75 s, peak speed 74.1 mph. No issues.

4 Exhibits (figures)

Figures are rendered from the 3D reconstruction at the time each observation is tested. Eye-level views use a 60° camera at the witness's marked position and eye height; the banner states the ray-cast result. Renders illustrate geometry only; they do not show what a person perceived.

No renders are available for this case yet.

5 Independent checks

These checks are deterministic geometry and physics computed from the reconstruction; they are not model output. Line of sight is a ray cast from the witness's eye to the target, repeated from 25 deterministic positions in a 2 m disc around the marked position (eye height ±0.1 m) to test sensitivity to where the witness stood. Perception-reaction time compares the time available against a 1.0–2.5 s range. Plausibility flags accelerations over 1.0 g outside ±0.5 s of impact, jumps over 5 m per 0.05 s and speeds over 60 m/s.

CHECK	RESULT	FINDING OF THE CHECK
Line of sight wit_A	CLEAR	Line of sight wit_A → veh_A at t = 2.00 s: **Clear** (target 639.5 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **2.4°** off the line of gaze (gaze assumed toward mark_impact); not peripheral (30° or less); camera FOV 60°. Basis: Witness A statement (fictional), modeled on NTSB/HAR-17/02 `p.1-2`, `p.29-30` witness account
Line of sight wit_B	BLOCKED	Line of sight wit_B → veh_A at t = 9.90 s: **Blocked by veh_T_trailer** (target 28.3 m away). Robustness: blocked from **25 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — robust (veh_T_trailer 15, veh_T_tractor 10). Gaze: target **28.5°** off the line of gaze (gaze assumed toward mark_impact); not peripheral (30° or less); camera FOV 60°. Basis: Deposition of Witness B (fictional), claim at 27:46 (repeated shortly after); position marked on Exhibit 4 at 27:04 while saying only 'I was over by the corner, near the shelter'
Line of sight wit_C	CLEAR	Line of sight wit_C → veh_T_tractor at t = 5.00 s: **Clear** (target 29.2 m away). Robustness: blocked from **0 of 25** positions within 2 m of the marked position (eye height ±0.1 m) — not blocked. Gaze: target **17.1°** off the line of gaze (gaze assumed toward mark_impact); not peripheral (30° or less); camera FOV 60°. Basis: Witness C statement (fictional); consistent with NTSB/HAR-17/02 `p.1-2` and `p.5`
Perception-reaction time veh_A	SUFFICIENT	Perception-reaction time: veh_A had **10.43 s** from first sight of veh_T_trailer to impact (first visible t = 0.00 s); PRT range 1.0–2.5 s, typical 1.5 s — **sufficient**. Time from the conflict first arising to impact: 2.88 s. Basis: meta.impactT
Perception-reaction time veh_T_tractor	SUFFICIENT	Perception-reaction time: veh_T_tractor had **10.40 s** from first sight of veh_A to impact (first visible t = 0.03 s, earlier view blocked by occ_crest); PRT range 1.0–2.5 s, typical 1.5 s — **sufficient**. Basis: meta.impactT
Plausibility veh_A	PASS	Kinematic plausibility: peak 0.02 g at t = 4.75 s, peak speed 74.1 mph. No issues. Basis: inferred: NTSB/HAR-17/02 `p.1` (910 ft post-impact), `p.2` & `p.15` (74 mph, no braking, headlights off), `p.5` (no pre-impact marks); dims: typical full-size sedan, assumed
Plausibility veh_T_tractor	PASS	Kinematic plausibility: peak 0.20 g at t = 4.20 s, peak speed 13.7 mph. No issues. Basis: NTSB/HAR-17/02 `p.1` (left turn from WB turn lane onto NE 140th Ct; stopped south of retail business), `p.1-2` (witness could not tell if stopped or moving very slowly)
Plausibility veh_T_trailer	PASS	Kinematic plausibility: peak 0.19 g at t = 5.70 s, peak speed 13.6 mph. No issues. Basis: NTSB/HAR-17/02 `p.6` (71 ft 7 in overall; impact 45 ft 5 in behind tractor front)

Skid-to-speed: not computed — the case contains no pre-impact tire marks to measure.

6 Validation against NTSB HAR-17/02

The reconstruction (world_v2.js) was measured against 12 facts stated in the published investigation. **11 of 12** are reproduced within tolerance. Facts not reproduced are reported, not hidden.

PUBLISHED FACT	PUBLISHED	MODEL	TOLERANCE	RESULT	CITATION
Car speed just before impact mean over t=9.39..10.34 s	74 mph	74 mph	±2 mph	MATCH	NTSB/HAR-17/02 `p.2`, `p.15` (facts.md F17)
No braking before impact (max deceleration in the 5 s before impact) over t=5.39..10.34 s, 0.5 s smoothing	0 g	0 g	±0.1 g	MATCH	NTSB/HAR-17/02 `p.15` (facts.md F18); `p.5` no pre-collision roadway marks (F22)
Time for the car to cover the 1,132 ft from the crest to impact (NTSB mutual sight time) car path starts only 343.4 m (10.39 s) before impact; cannot cover 345 m	10.40 s	not reproduced	±0.3 s	NO MATCH	NTSB/HAR-17/02 `p.29` (facts.md F28)
Vertical crest west of the intersection (sight restriction) - distance from impact point to the crest occluder's near edge impact point to the occluder edge nearest the intersection, along the car's travel direction	345 m	346 m	±15 m	MATCH	NTSB/HAR-17/02 `p.22-23`, `p.29` (facts.md F10, F27): 1,132 ft
Impact in the right (outside) eastbound lane at the NE 140th Ct crossing (car front at impact vs. case-frame POI) car front at t=10.39s	(0.00, 0.00) m	(-1.32, 0.00) m	x ±5, z ±1.83 m	MATCH	NTSB/HAR-17/02 `p.3`, `p.5` (facts.md F21); lane 12 ft (F6) -> +/-1.83 m lateral; origin = POI per case.json coordinate_frame_note (modeling convention)
Impact point on the trailer, distance behind the front of the tractor at t=10.39s	13.84 m	13.60 m	±1 m	MATCH	NTSB/HAR-17/02 `p.6` fn.6 (facts.md F16): 45 ft 5 in; tolerance = half the car's width
Tractor-semitrailer combination length at t=9.25s (tractor/trailer 0 deg apart)	21.82 m	21.82 m	±0.5 m	MATCH	NTSB/HAR-17/02 `p.1`, `p.6` (facts.md F15): 71 ft 7 in
Time of day of the collision	16:36:12.7	16:36	±60 s	MATCH	NTSB/HAR-17/02 `p.1`, `p.15` (facts.md F1): car data recorded collision at 4:36:12.7 p.m.
Pole strike (airbag deployment) time after first impact	8.40 s	8.42 s	±0.5 s	MATCH	NTSB/HAR-17/02 `p.5` (facts.md F26); pole struck on post-impact path `p.1` (F24)

PUBLISHED FACT	PUBLISHED	MODEL	TOLERANCE	RESULT	CITATION
Car travel from point of impact to final rest straight line, impact point to car centre at end	277 m	279 m	±20 m	MATCH	NTSB/HAR-17/02 `p.1`, `p.3` (facts.md F25; police_report `p.3` 'POI to final rest'): about 910 ft
Final rest off the road to the right (south) of the eastbound lanes	>= 3.35	23.90 m	±0 m	MATCH	NTSB/HAR-17/02 `p.1`, `p.3` (facts.md F23, F25: left the right roadside, rest in a front yard); 12-ft lane half-width + 5-ft right shoulder (F6) = 3.35 m from lane centre
Final rest perpendicular to the highway	90 deg	90 deg	±20 deg	MATCH	NTSB/HAR-17/02 `p.1`, `p.3` (facts.md F25)

7 Methodology

1. **Ingest.** Each evidence file is stored unchanged; text is extracted from documents and the deposition transcript is kept with its timecodes so that every statement can be cited as `Depo MM:SS`.
2. **Reconstruction.** A multimodal model (gemini-4-argon) reads the evidence and writes a time-parameterised 3D scene: road geometry, vehicles and their paths, occluders, and each witness's marked position, eye height and observation time. Every object and value carries a citation to the evidence it came from; values without direct support are labelled *inferred* or *assumed*.
3. **Review rounds.** The scene is rendered from overhead and from each witness's eye position. The model reviews its own renders against the evidence and revises the scene; this case went through 2 rounds. Each version is recorded with its inputs and prompt (Section 8).
4. **Independent checks.** Deterministic code, separate from the model, ray-casts every witness's sight line (with a 25-position robustness test), measures how far the target was from the witness's gaze, computes perception-reaction time available, skid-to-speed ranges where marks exist (μ 0.6–0.8), and flags physically implausible motion.
5. **Validation.** Where a published investigation exists, the reconstruction is measured against its stated facts (Section 6).
6. **Findings.** Each finding states a status (contradiction, consistent, unverifiable), cites the testimony and documents it rests on, and lists the independent checks that bear on it.

LIMITATIONS

Kinematic reconstruction: motion fitted to cited evidence; not a crash-physics simulation. Sight lines are geometric: a clear line of sight does not establish that a witness noticed an object, and gaze direction is assumed where testimony does not state it. Results depend on the marked witness position, which the robustness test brackets to 2 m. Conclusions must be confirmed by a qualified reconstructionist before any use.

8 Provenance

VERSION	MODE	MODEL / PROMPT	CREATED	OUTPUT SHA-256
world_v1 v1	mock	gemini-4-argon `argon/prompts/generate_world_v2.md @ 93c75e220362`	October 25, 2026 at 12:21 PM PDT	`ef838b81682f0558`
world_v2 v2 ← world_v1	mock	gemini-4-argon `argon/prompts/critique.md @ fe363a569e50`	October 25, 2026 at 12:23 PM PDT	`d5924474a55a15a1`

INPUTS TO WORLD_V2 (SHA-256)

FILE	BYTES	SHA-256
`text/deposition.txt`	30,738	`4c0cb2f76681059939846bdd46483ce67d918787c4bde6676618a251f2ef8263`
`text/deposition_transcript.txt`	20,708	`416d65e03cfb46aed00d9b2caeea511a145013a23f69bc6e304b6bba2e100432`
`text/facts.txt`	6,883	`16a2ac328322655c10775315a2666ebef5eb8a76ddab222b6e7e8c98fcf7cfad`
`text/police_report.txt`	6,532	`6b994577e18a90415850198d288091c0e19de0ee2ce655d11072e472e0c85791`

FILE	BYTES	SHA-256
text/witness_A.txt	1,729	57b80e791b7642498bbfb3756d46ecf878beeb5cee317c2e04313c0fc04e5c8c
text/witness_C.txt	1,507	14873e9544ba08b9b7e61a3b3fb368ca313b76ec2ff98bc434cd73ee8cd8656e
evidence/deposition.mp4	4,244,843	8811f02e4e9e62da6a70a698186374a76ebf5ace7abb39016df9345a318168fd
evidence/map.png	54,576	e44f99efb5dd48300e638c39b92ad9a70c2642c31da7f66dd30b9248bf251aaa

Reconstruction file: /cases/demo/worlds/world_v2.js · SHA-256 d5924474a55a15a1b2ae29c1a5a99bd6a61f2cb8b900a6c69018a754ab2427d7. Findings and checks last updated October 25, 2026 at 12:25 PM PDT.

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Reviewed by (reconstruction expert)

Date